

BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Semester Final Exam
 Duration: 1 Hour 40 Minutes

Semester: Fall 2022
 Full Marks: 30

CSE 221: Algorithms

Answer the following questions.
 Figures in the right margin indicate marks.

Name:	ID:	Section:
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- 1 a. C03** As we all know, the Spanish National Football Team is notorious for their passing football, especially their back passes. After their disastrous 2022 World Cup campaign, The King of Spain made you the coach and declared that no back passes are allowed on the field from now on till Spain scores 2 goals, that is, Simon the goalkeeper can only pass to the defenders Rodri and Cesar, but Rodri and Cesar cannot pass the ball back to Simon. Also, every pass now comes with a cost– and the more cost players accumulate, the angrier the King gets. These are the costs of the passes: **06**

Passes	Cost
Simon → Rodri	2
Simon → Cesar	4
Rodri → Cesar	1
Rodri → Pedri	7
Cesar → Gavi	3
Gavi → Pedri	2
Pedri → Morata	1
Gavi → Morata	5

Find the lowest cost for the ball to reach every player starting from Simon, using any algorithm you find useful. The King will need to see every single step of the algorithm, or he will not believe you and fire you on the spot.

- b. C03** After scoring 2 goals, Spain changed their strategy and started to back pass again incorporating 2 more players, the master of dark arts Busquets and the young sensation Ansu Fati. Now they are no longer concerned with the cost of the pass as well. These are the passes for the new strategy: **04**

Passes
Simon → Rodri
Simon → Cesar
Rodri → Busquets
Rodri → Pedri
Cesar → Rodri
Busquets → Gavi
Busquets → Pedri
Gavi → Cesar
Gavi → Pedri
Gavi → Fati
Fati → Morata
Pedri → Fati
Morata → Pedri

Now using a suitable algorithm, **find** out the largest group of players who can pass the ball among themselves. For example, one such group can be (Fati, Morata, Pedri) where Morata can pass to Pedri, Pedri can pass to Fati and Fati can pass to Morata. It is important to keep in mind that the king will be observing every step of the algorithm. Any discrepancies can lead you to lose your job.

- 2 a.** You are given the following table containing symbols and their frequencies:

CO2

Symbol	A	B	C	D	+
Frequency	40	10	20	15	15

- Build** the Huffman code tree and find the codewords for each character.
- Decode** 100010111001010 using the Huffman code that you generated.

**04
01**

- b.** You are given the arrival and departure times of eight trains for a railway platform, each in the following format: *[arrival time, departure time]*. Only one train can use the platform at a time.

CO4

Suppose, you have got the following train-use requests for the next day.

{ [8, 13), [6, 9), [11, 14), [2, 7), [1, 7), [12, 20), [7, 13), [13, 20) }

- Find** the maximum number of trains that can use the platform without any collision.
- Determine** the minimum number of platforms that needs to be there to ensure the arrival and departure of all these trains without collision.

**03
02**

3 Suppose you went to Miniso to buy a Teddy Bear Plushie (a soft toy or doll) as a gift for your younger sister's birthday. The plushie is 700 BDT. At Miniso, all items are subject to a 6% additional VAT charge.

You bought one plushie and gave the cashier 750 BDT. The cashier has a huge supply of 1 taka, 2 taka, and 5 taka coins in the cashbox. You don't want to carry many coins, so you asked her to return the change using a minimum number of coins.

- | | | | |
|------------|------|---|-----------|
| C04 | i) | Using the Dynamic Programming approach, determine how many coins she should return in this scenario. | 06 |
| C04 | ii) | Which coins did you get from the cashier? Show how your table was used to pick up these coins. You may use arrows and circles to point to the chosen cells. | 02 |
| C07 | iii) | Suppose you did not apply memoization to find the answer for question (i). What will be the time complexity of that brute force approach? Explain with proper reasoning. | 02 |

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Morata → Fati
Rodri → Pedri
Rodri → Cesar

Now using a suitable algorithm, **find** out the largest group of players who can pass the ball among themselves. For example, one such group can be (Fati, Morata, Pedri) where Morata can pass to Pedri, Pedri can pass to Fati and Fati can pass to Morata. It is important to keep in mind that the king will be observing every step of the algorithm. Any discrepancies can lead you to lose your job.

- 2 a.** You are given the following table containing symbols and their frequencies:

CO2

Symbol	A	B	C	D	#
Frequency	35	15	25	10	20

- Build** the Huffman code tree and find the codewords for each character.
- Decode** 100010111001010 using the Huffman code that you generated.

**04
01**

- b.** You are given the arrival and departure times of eight trains for a railway platform, each in the following format: *[arrival time, departure time]*. Only one train can use the platform at a time.

CO4

Suppose, you have got the following train-use requests for the next day.

{ [8, 12), [6, 9), [11, 14), [2, 7), [1, 7), [12, 20), [7, 12), [13, 19) }

- Find** the maximum number of trains that can use the platform without any collision.
- Determine** the minimum number of platforms that needs to be there to ensure the arrival and departure of all these trains without collision.

**03
02**

3

A team of two infamous thieves, Denver and Nairobi, planned to rob the famous Louvre Museum. Before the scene, they both agreed on the fact that none of them will break any item as all the items in the Louvre are too precious, and taking a fraction of any item won't sell in the black market. If it fits in the bag as a whole, they will take it, otherwise, leave it as it is.

Both of them arrived at the Louvre with an empty knapsack weighing a total of 8 kg. Despite the fact that both thieves are experts in their fields, they take slightly different approaches. Denver believes he will use a Dynamic Programming Approach to rob the items in the most efficient manner possible. Nairobi, on the other hand, believes that if she chooses a Greedy Approach, she will make the most money.

The objects in the Louvre Museum are listed below.

Objects	Jewelry	Sculpture	Painting	Book	Mummy
Profit (\$)	5	9	5	4	6
Weight (Kg)	3	5	4	1	12

- C04** i) **Calculate** the maximum profit Denver can make using his strategy. What items did he pick up? **Show** how Denver used the DP table to select these objects. You may use arrows and circles to point to the chosen cells. **08**
- C04** ii) Does Nairobi's belief remain valid after the robbery? **Prove** it. **02**

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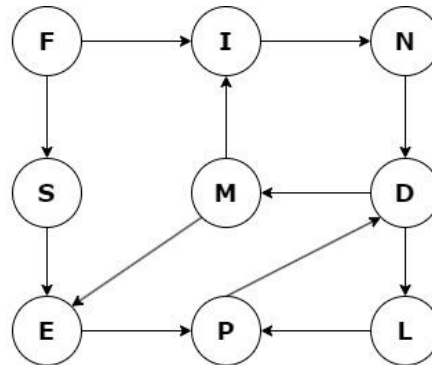
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CSE 221: Algorithms

Answer all of the following questions in your script.
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1. Sherlock Holmes found himself in a dire situation where James Moriarty imprisoned his assistant Watson in a maze. The maze is made of 9 different rooms. Each room has a part of the key with an alphabet written on it. Moriarty tells Sherlock that only if he can attach all the key chain parts in exactly this order: “**F-I-N-D-M-E-P-L-S**”, then he can unlock the room with that key where Watson is kept. The map of the maze is given below:



- a) [CO1] Now you know something about a graph traversal algorithm that travels from the source to as far as possible along each branch (depth-wise traversal). Simulate/Draw the traversal and **fill up** the following table (in your script) with the start and ending time for each node you achieved while doing this task. Keep in mind, wherever you have to choose between different neighbors to explore, follow the sequence that will make the key chain parts in the sequence that Sherlock needs.

06

Node	D	E	F	I	M	S	L	P	N
Start									
End									

- b) [CO1] When Sherlock tries to save his friend, water suddenly fills the rooms. He needs to find a way to contain the water somehow so that the water doesn't overflow from one place to another. So, he created different isolated closed groups of rooms where he can travel anywhere in that particular group, meaning they have a path from any room to any other room inside the group. After doing so, he finds out that he can make a few of these groups of rooms inside this maze. A group can contain 1 to N number of rooms where N is the total number of vertices, and no two groups can share the same room.

- i) **Mention** the name of a suitable algorithm for this task.
- ii) **Show** simulation for the graph given above.

01
03

You can reuse/refer to your answer from question (a) to reduce work.

2. a) [CO2] “Dijkstra’s Algorithm can find the single source shortest paths for any graph.” 02
Do you think this statement is true? If yes, **state** your reasoning. Else, **validate** it to be false with an example graph.

- b) [CO1] Imagine a daring explorer on a mission to uncover hidden treasures scattered across a cluster of islands. The listed islands have a bidirectional bridge between them: 03

Aqua	Blaze	Coral	Dune	Coral	Flare
Aqua	Coral	Blaze	Ember	Ember	Flare
Blaze	Coral	Dune	Ember	Ember	Glimmer
Blaze	Dune	Dune	Flare	Flare	Glimmer

If each bridge has a travel time of 5 minutes, give an $O(V+E)$ algorithm that can find the shortest travel time from Aqua to every other island.

Explain your idea in an executable code/pseudocode/flow-chart format.

- c) [CO1] Continuing on the previous question, suppose the travel time of each bridge is not equal anymore. The updated travel times are as follows:

Bridge between		Travel time	Bridge between		Travel time
Aqua	Blaze	5	Dune	Ember	6
Aqua	Coral	13	Dune	Flare	6
Blaze	Coral	4	Coral	Flare	16
Blaze	Dune	6	Ember	Flare	3
Coral	Dune	5	Ember	Glimmer	11
Blaze	Ember	7	Flare	Glimmer	2

Now, the explorer wants to find a set of bridges that connects all the islands with the **minimum total travel time**. However, the bridge between Ember and Flare and the bridge between Flare and Glimmer suddenly become vulnerable and may break down anytime. So, the explorer decides to avoid these two bridges while constructing the subgraph that connects all islands.

Answer the following questions based on this scenario.

- i) Which bridges will the explorer choose?
ii) What will be the total travel time?

Show your works on the answer script.

03
02

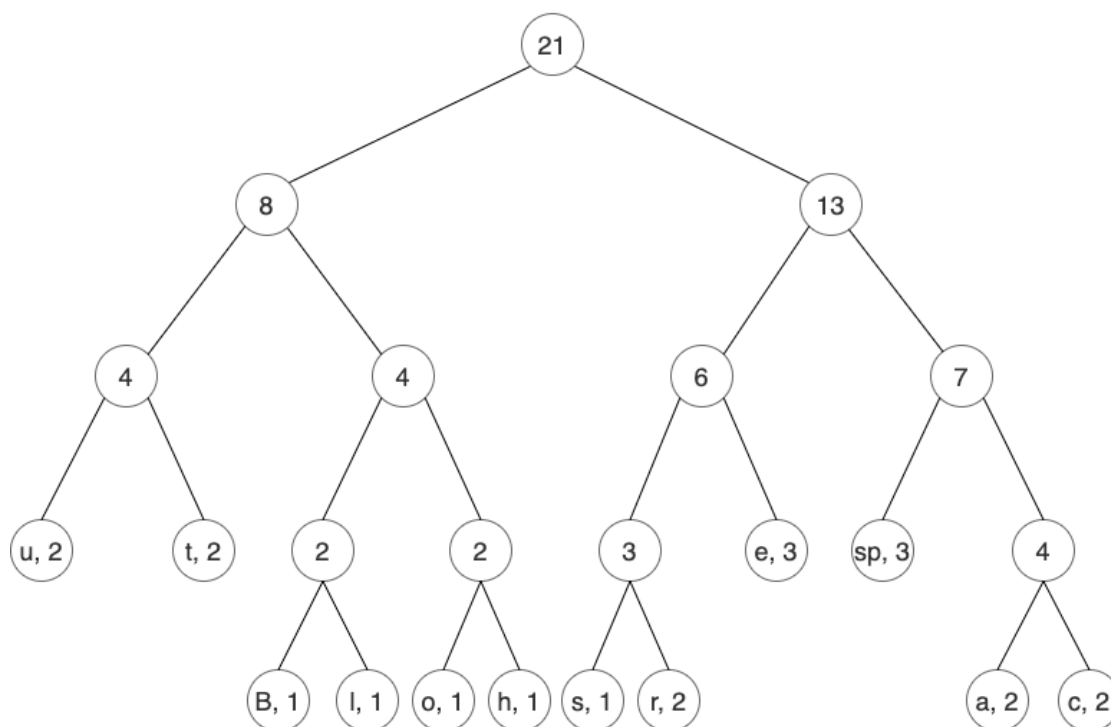
3. a) [CO1] Welcome to the exciting realm of Bracula Gaming World, where you can earn lots of money by playing different games. But there's a twist – time is limited to just 20 hours! Each game has its own time limit and prize money if played for the full duration. However, you can play for a **fraction of the time** and still earn money in proportion. Check the list of games below.
Your task is to make the most of your 20 hours and collect **as much money as possible**.

Game Name	Total Time Limit	Prize Money For Full Duration
Road Rush	15 hours	75\$
GTA Vice City	20 hours	90\$
Virtual Cop	3 hours	30\$
House of the Dead	6 hours	55\$
Far Cry	2 hours	17\$
Devil May Cry	10 hours	47\$

- Name the algorithm tailored for this situation.
- Write the names of the games and their respective playing times that will lead to maximum prize money. Show a simulation of your algorithm.

01
04

- [CO1] You are given the following tree generated by simulating the Huffman encoding algorithm on a text message.



- Write the encoding for each of the characters. Consider left to be 0, and right to be 1 when traversing the tree.
- Decode the following encoded message using the character codes you wrote on question (i).
01001001111011110000101111011000101101100010111101110100110110001111000101

04

01

Please Turn Over

4. DNAs contain genetic and hereditary information of almost every living organism. DNA Sequence Matching is a procedure to compare two DNA sequences and find similarity between them. Given below is the recurrence relation of a dynamic programming algorithm to perform sequence matching of two DNAs.

$$seq(i, j) = \max \begin{cases} seq(i, j-1) - 1, \\ seq(i-1, j) + p(i, j), \\ seq(i-1, j) - 1 \end{cases}$$

where,

$$\begin{aligned} seq(i, j) &= 0, & \text{if } i = 0 \text{ or } j = 0 \\ p(i, j) &= 1, & \text{if } DNA1[i] = DNA2[j] \\ p(i, j) &= -2, & \text{otherwise} \end{aligned}$$

As you can see, the formula takes two indices, i and j of two DNA sequences given as input. Then it produces a similarity score as output. We want to apply this formula to compare the following DNA sequences.

index	1	2	3	4	5
DNA1	G	C	G	T	A
DNA2	C	T	G	A	G

Now answer the following questions:

- [CO1] Put appropriate values to **fill the gaps** in the line below (answer in your script): **02**
 $seq(_, _)$ represents the final similarity score of the given DNAs.
- [CO1] **Apply** the formula to calculate a similarity score between the DNA sequences given above. Show your work with either a recursion tree or a memory table. **05**
- [CO2] **State** the time complexity of this algorithm with proper reasoning. **02**
 You can assume that, the lengths of the DNA sequences are N and M respectively.
- [CO3] Mriaslun, a talented algorithmist, claims that he can implement the algorithm in an optimized way so that it only takes $O(\min(N, M))$ space. Can you do it too? **Explain** your idea. **01**

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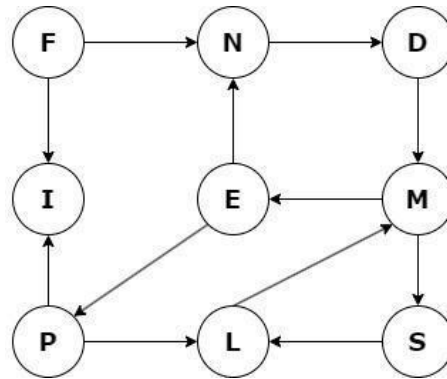
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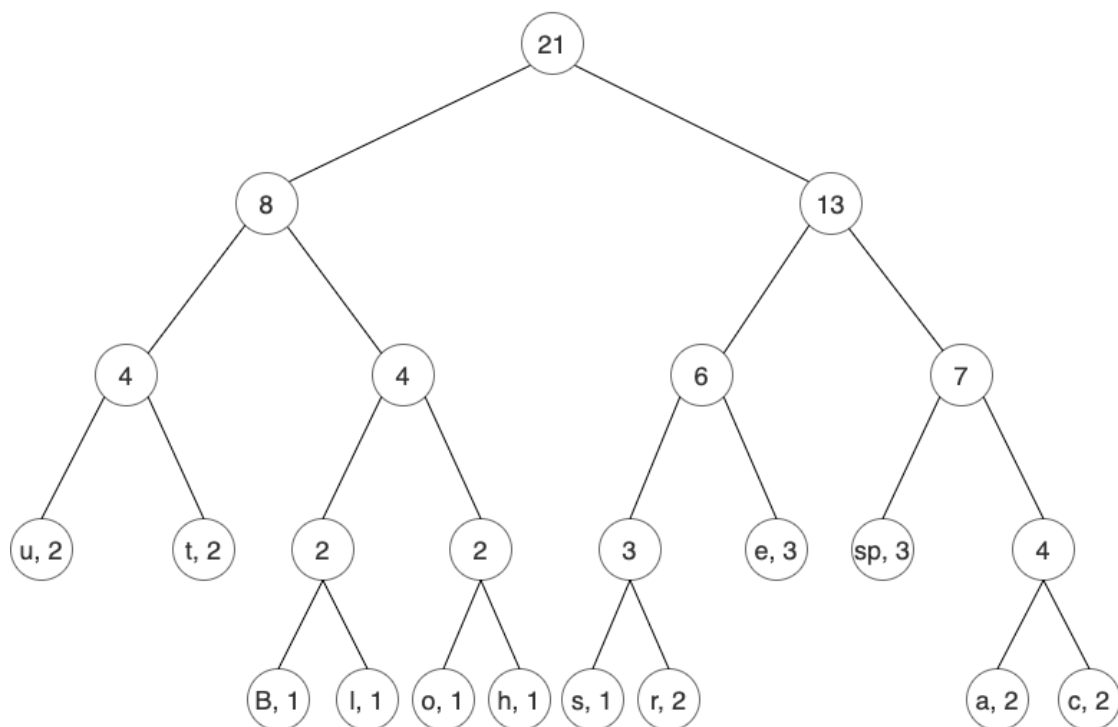
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GTA Vice City	20 hours	95\$
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Far Cry	3 hours	18\$
Devil May Cry	11 hours	50\$

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01
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- 1** You have just joined a new university and are exploring the campus, which consists of several buildings connected by one-way bridges. These bridges form a directed graph, where each node represents a building and each edge represents a bridge between two buildings. You aim to analyze how the buildings and bridges are connected to understand the campus layout better and improve your navigation between classes and facilities.

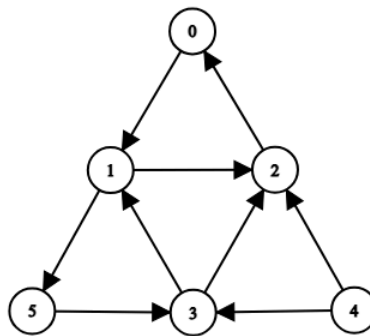


Figure 1

- CO2** a) **Mention** which graph representation will let you check whether any two buildings have a bridge between them in $O(1)$ time. **01**
- b) From Figure 1
- CO1** i. **Simulate** an algorithm to find the Strongly Connected Components (SCC). **04**
- CO3** ii. Suppose a bridge gets blocked. **Explain** how you can efficiently calculate the updated SCCs from b(i) without rerunning the algorithm on the entire graph. **02**
- CO3** c) A building is called “*Accessible*” if it's possible to start from there, visit some (or all) of the other buildings, and eventually return to that building. **03**

Propose an algorithm with a pseudocode/programmable code/step-by-step logical instructions to determine whether every building on the campus is *Accessible*. The algorithm should run in **at most** $O(V \times (V + E))$ time.

2

Imagine, you and your best friend are trapped in a dungeon, trying to escape quickly. The dungeon is made up of chambers connected by two-way tunnels. However, these tunnels may contain dangers that drain your energy fighting it. You have a map showing how much energy you will require while passing through each of the tunnels. The energy requirements for each tunnel are not necessarily the same.

Now answer the following questions.

- a. **CO1** You are given the map as an Adjacency List where the chambers are numbered from 1 to N. You are in chamber 1 and aiming to reach chamber N using a path that requires minimum energy. **02**

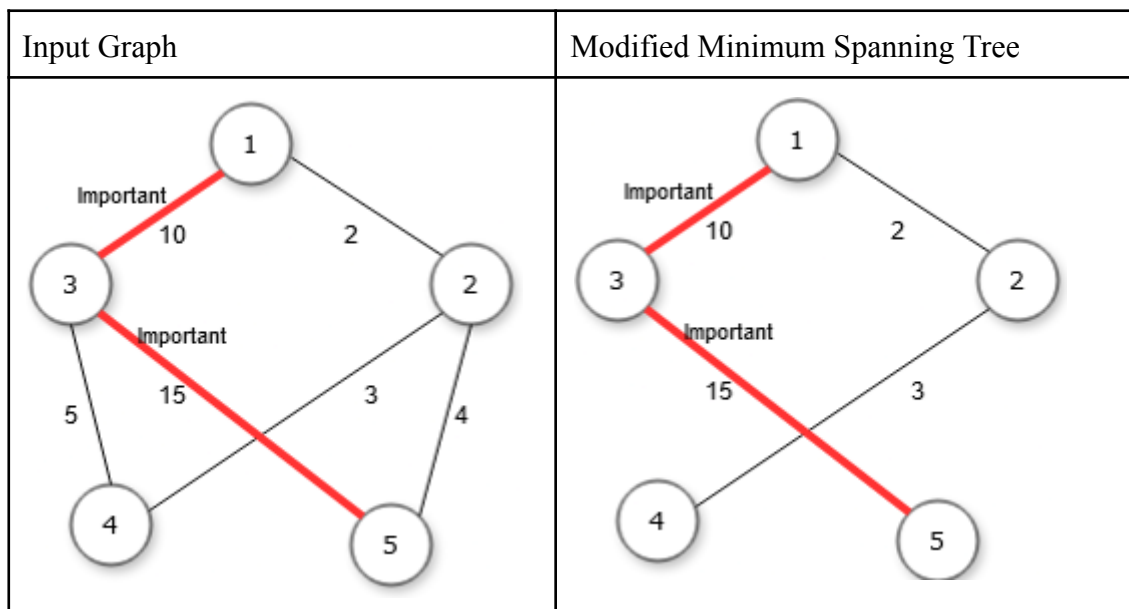
Name the algorithm you prefer to solve this problem. **Explain** your choice.

- b. **CO2** If some tunnels contain an energy drink that will *restore* your energy by a certain amount, will your proposed algorithm in (a) work? **Explain** Briefly. **01**

- c. **CO2** You realize that you forgot some of your belongings in some tunnels. Hence, you want to construct the MST such that those tunnels are always included no matter how dangerous they are. **04**

Formally, you have an undirected weighted graph with some edges marked as “*important*” and you need to propose a modification to Kruskal’s algorithm such that you can find the MST that always includes those important edges (even if it increases the total cost).

For simplicity, you can assume the important edges/tunnels do not create any cycles. Look at the example below.



Present your algorithm with a programmable code/ pseudocode/ step-by-step instruction.

- d. **CO3** The map of the island is too big to carry. Thus, your friend came up with an idea to use Kruskal's Algorithm to only store the Minimum Spanning Tree (MST) of the graph. He stated that *“The minimum energy needed to travel between any two chambers in the MST is the same as in the Original Map”*. **03**
- Do you agree with the statement? **Justify** your answer with proper examples and necessary details.

3 Consider the following scenario

You are designing a data compression system for a critical database server. You have analyzed the following character frequencies from the database logs:

Character	Frequency
A	45
B	13
C	12
D	16
E	9
F	9
G	12

Now answer the following questions:

- a. **CO1** **Construct** a Huffman tree for the given characters and their frequencies. Additionally, calculate the space saved compared to using fixed 3-bit encoding for each character. Show your work step by step. **04**
- b. **CO2** **Write** the Huffman Code for each character. **02**
- c. **CO2** **Explain** how greedy property is used in Huffman coding and how it provides an optimal code. **02**
- d. **CO2** Two friends are tasked with building the given Huffman tree. While both friends use priority queues in the process, **02**
- friend 1 inserts the newly created nodes at the end of the same frequency nodes.
 - friend 2 inserts the newly created node at a random position between the same frequency nodes.

Explain which technique will provide more optimal code and why. If they are bound to provide equally optimal code, **explain** why.

4 Answer the following questions:

a. Can merge sort be written as a dynamic programming algorithm? **Explain** your answer. **02**
CO2

b. We are calculating the longest common subsequence (LCS) between two strings, $S = X_1X_2X_3X_4$ and $T = Y_1Y_2Y_3Y_4$. To do this, we fill an array C where $C_{i,j}$ represents the length of the LCS between the prefix of length i from S and the prefix of length j from T .

Some of the entries in the array C are currently masked using different symbols other than integers as shown below.

	Y_0	Y_1	Y_2	Y_3	Y_4
X_0	0	0	0	0	0
X_1	0	0	0	0	1
X_2	0	0	0	1	1
X_3	0	0	1	1	♡
X_4	0	1	♣	♠	◇

- CO1 i. What does the cell $C_{0,0}$ represent? **01**
 ii. What can be said about the characters X_1 and Y_4 ? **01**
 iii. What can be said about the characters X_2 and Y_4 ? **01**
 iv. What is the value of ♡? **01**
 v. What is the value of ♣? **01**
 vi. When we are filling up the i -th row of our dynamic programming table C , what rows do we need to have access to? **01**
 CO3 vii. Given the observation above can we optimize our space complexity further? **02**

BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Final Exam
Duration: 1 Hour 40 Minutes

Semester: Fall 2024
Full Marks: 40

CSE 221: Algorithms

Answer the following questions.
Figures in the right margin indicate marks.

Name:	ID:	Section:
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- 1** You have just joined a new university and are exploring the campus, which consists of several buildings connected by one-way bridges. These bridges form a directed graph, where each node represents a building and each edge represents a bridge between two buildings. You aim to analyze how the buildings and bridges are connected to understand the campus layout better and improve your navigation between classes and facilities.

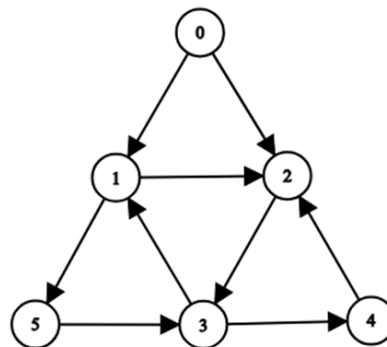


Figure 1

- CO2** a) **Mention** which graph representation will let you check whether any two buildings have a bridge between them in $O(1)$ time. **01**
- b) From Figure 1
- CO1** i. **Simulate** an algorithm to find the Strongly Connected Components (SCC). **04**
- CO3** ii. Suppose a bridge gets blocked. **Explain** how you can efficiently calculate the updated SCCs from b(i) without rerunning the algorithm on the entire graph. **02**
- CO3** c) A building is called “*Accessible*” if it's possible to start from there, visit some (or all) of the other buildings, and eventually return to that building. **03**

Propose an algorithm with a pseudocode/programmable code/step-by-step logical instructions to determine whether every building on the campus is *Accessible*. The algorithm should run in **at most** $O(V \times (V + E))$ time.

2

Captain Jack Sparrow has arrived at a treasure island, planning to steal the treasure quickly. The island has two-way pathways, each with hidden dangers. To handle these dangers, Jack brought a weapon. Facing danger in a pathway requires using some bullets. He has a map of the island showing the number of bullets needed to fight the dangers in the pathways. Bullet requirements for each pathway are not necessarily the same.

Now, answer the following questions.

- a. Jack has been given the map as an Adjacency List where the locations are numbered from 1 to N. Jack is in chamber 1 and aiming to reach location N using a path that requires minimum bullets. 02
- CO1

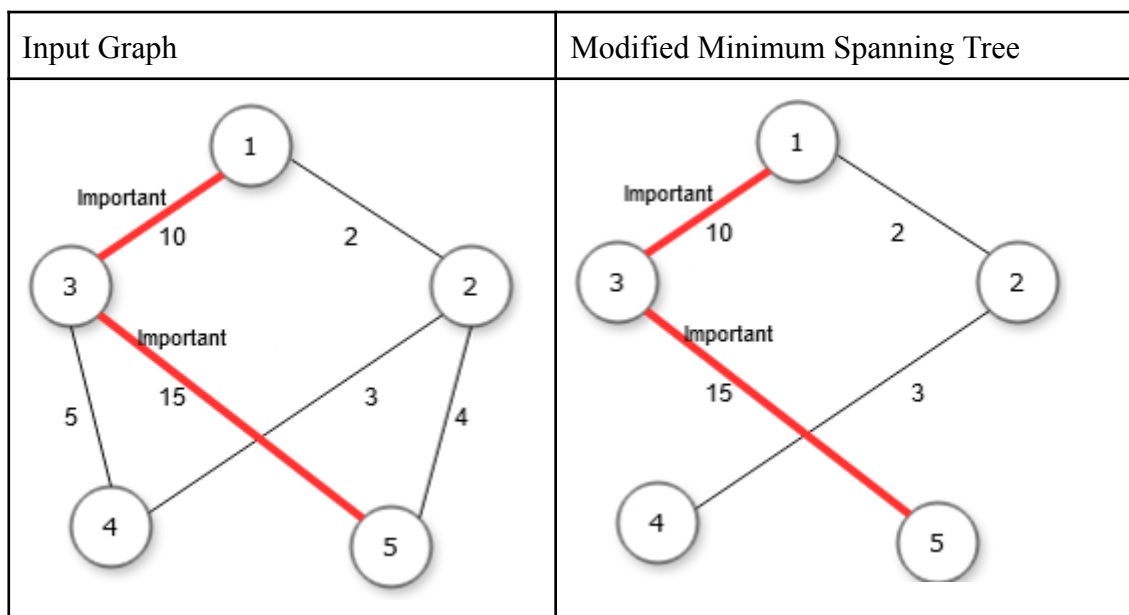
Name the algorithm you prefer to solve this problem. Explain your choice.

- b. If some pathways contain hidden supplies of bullets that will refill his ammo by a certain amount, will your proposed algorithm in (a) work? Explain Briefly. 01
- CO2

- c. Jack realizes that he left some of his belongings in some pathways. Hence, he wants to construct the MST such that those tunnels are always included no matter how dangerous they are. 04
- CO2

Formally, he has an undirected weighted graph with some edges marked as “important” and he needs to propose a modification to Kruskal’s algorithm such that he can find the MST that always includes those important edges (even if it increases the total cost).

For simplicity, you can assume the important edges/tunnels do not create any cycles. Look at the example below.



Present your algorithm with a programmable code/ pseudocode/ step-by-step instruction.

- d. **CO3** The map of the island is too big to carry. Thus, your friend came up with an idea to use Kruskal's Algorithm to only store the Minimum Spanning Tree (MST) of the graph. He stated that “*The minimum energy needed to travel between any two chambers in the MST is the same as in the Original Map*”. **03**

Do you agree with the statement? **Justify** your answer with proper examples and necessary details.

3 Consider the following scenario

You are designing a data compression system for a critical database server. You have analyzed the following character frequencies from the database logs:

Character	Frequency
A	50
B	15
C	20
D	10
E	5
F	5
G	30

Now answer the following questions:

- a. **CO1** **Construct** a Huffman tree for the given characters and their frequencies. Additionally, calculate the space saved compared to using fixed 3-bit encoding for each character. Show your work step by step. **04**
- b. **CO1** **Write** the Huffman Code for each character. **02**
- c. **CO2** **Explain** how greedy property is used in Huffman coding and how it provides an optimal code. **02**
- d. **CO2** Two friends are tasked with building the given Huffman tree. While both friends use priority queues in the process, **02**
- friend 1 inserts the newly created nodes at the end of the same frequency nodes.
 - friend 2 inserts the newly created node at a random position between the same frequency nodes.

Explain which technique will provide more optimal code and why. If they are bound to provide equally optimal code, **explain** why.

4 Answer the following questions:

a. Can merge sort be written as a dynamic programming algorithm? **Explain** your answer. **02**
CO2

b. We are calculating the longest common subsequence (LCS) between two strings, $S = X_1X_2X_3X_4$ and $T = Y_1Y_2Y_3Y_4$. To do this, we fill an array C where $C_{i,j}$ represents the length of the LCS between the prefix of length i from S and the prefix of length j from T .

Some of the entries in the array C are currently masked using different symbols other than integers as shown below.

	Y_0	Y_1	Y_2	Y_3	Y_4
X_0	0	0	0	0	0
X_1	0	0	0	0	1
X_2	0	0	0	1	1
X_3	0	0	1	1	♥
X_4	0	1	♣	♠	◇

- CO1 i. What does the cell $C_{0,1}$ represent? **01**
ii. What can be said about the characters X_2 and Y_3 ? **01**
iii. What can be said about the characters X_3 and Y_3 ? **01**
iv. What is the value of ♣? **01**
v. What is the value of ♠? **01**
vi. When we are filling up the i -th column of our dynamic programming table C , what columns do we need to have access to? **01**
CO3 vii. Given the observation above can we optimize our space complexity further? **02**

BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Semester Final Exam
 Duration: 1 Hour 50 Minutes

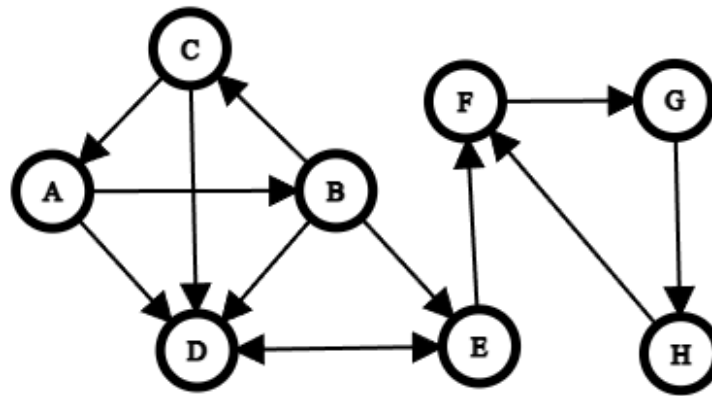
Semester: Fall 2025
 Full Marks: 35

CSE 221: Algorithms

Answer the following questions.
 Figures in the right margin indicate marks.

Name:	ID:	Section:
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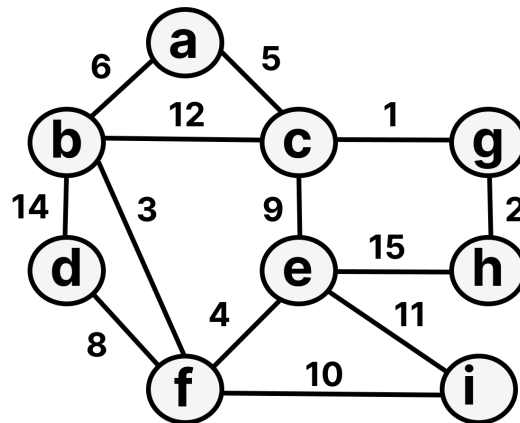
1 Consider the directed graph given below.



- a. **Simulate** a DFS on the graph to find the Discovery and Finish times for each vertex. **3**
CO1 Start with Vertex 'A'.
- b. Now, **simulate** an appropriate algorithm to identify the Strongly Connected Components. **3**
CO1 You can reuse your answer to Question (a) to save time.
- c. **Find** out the minimum number of edges to remove from the graph, so that there exists a **2**
CO1 topological order among the vertices. **Explain** your reasoning.
- d. If we consider each SCC [that you found in Question (b)] as a single vertex, will there be **2**
CO1 any remaining cycles in the graph? **Draw** the graph to justify your answer.

2

A country has several cities that need to be connected by highways. Each edge in the graph represents a potential highway connection between two cities, with weights indicating the **construction cost** of building that highway.



- a. The government wants to ensure all cities remain connected while minimizing the total construction cost. Simulate a suitable algorithm to determine which highways should be built. Report the total minimum construction cost. 4
- CO1
- b. Is it possible for a road network (not necessarily the one given above) to have multiple different configurations of highways, each resulting in the same minimum total construction cost? 2
- CO1

If yes, provide a simple example graph where this occurs. If not, justify your answer with proper reasoning.

- c. Now, assume **all highways have been constructed** as shown in the graph. The government has installed toll plazas on each highway. The edge weights now represent the toll fees for using each highway. You're planning a trip from a source city to a destination city and want to minimize your total toll cost. You have a special coupon that allows you to skip paying the toll on **only one** highway (you can use it at most once during your journey). 4
- CO2

Your friend proposed a modified approach: *First, apply standard Dijkstra's algorithm to find the shortest path from source to destination while keeping track of which edge has the highest toll cost on this path. Then, use the coupon on that edge to skip its toll cost. According to your friend, this will guarantee the minimum total toll cost from source to destination.*

Do you agree that this approach will guarantee the shortest path in all cases, regardless of how the cities are connected, given that all toll fees are positive? If yes, provide proper reasoning. If not, provide a counterexample showing where this approach will fail.

3

You are designing a lossless data compression module for a real-time monitoring system. After analyzing system logs, the following character frequency distribution is obtained:

Character	Frequency
D	5
A	30
R	12
B	18
C	10
E	4
U	8

- a.** Even though all character frequencies are unique, one can obtain different Huffman Trees for above characters. Is this possible? **Explain** whether the bit patterns and the optimality of the code will be affected by this. **2**
- CO1**
- b.** **Simulate** the Huffman Encoding algorithm for the given characters. Construct the huffman tree, generate Huffman code and bits required for each character. (consider left edge as 0, right as 1). **4**
- CO1**
- c.** If you were to send the message “BRACU” with an 8-bit encoding scheme, how many extra bits would have been required compared to Huffman encoding? Ignore the size of the code table. **2**
- CO2**
- d.** “In Huffman Encoding, a higher-frequency symbol always gets a shorter codeword than a lower-frequency symbol.” Is this statement always true? **Explain**. **2**
- CO2**

This is the LCS Table for two strings $X=X_1X_2X_3X_4$ and $Y=Y_1Y_2Y_3Y_4$

i\j	-	X_1	X_2	X_3	X_4
-	0	0	0	0	0
Y_1	0	0	0	0	0
Y_2	0	0	0	<u>1</u>	1
Y_3	0	0	1	1	2
Y_4	0	1	1	<u>1</u>	2

- a. What is the direction of cells where values are underlined, i.e which adjacent cells they came from? 2
CO1
- b. If String $Y = \langle \text{Rightmost four digits of Your Student ID} \rangle$, **construct** a correct String X . 2
CO1
- c. The formula to find LCS can be implemented both iteratively and recursively. Which one would be more efficient? **Explain** your answer briefly. 1
CO2

BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Semester Final Exam
 Duration: 1 Hour 50 Minutes

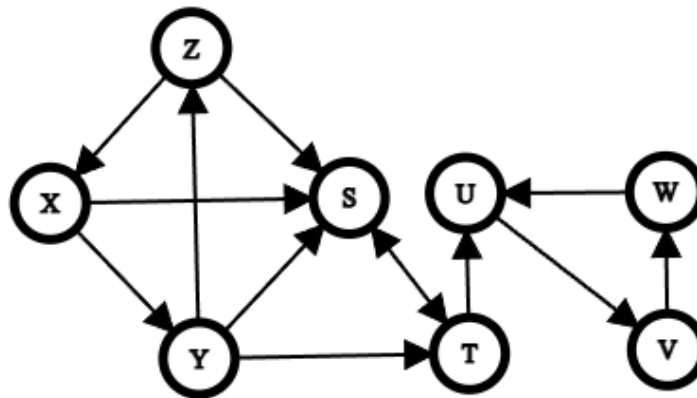
Semester: Fall 2025
 Full Marks: 35

CSE 221: Algorithms

Answer the following questions.
 Figures in the right margin indicate marks.

Name:	ID:	Section:
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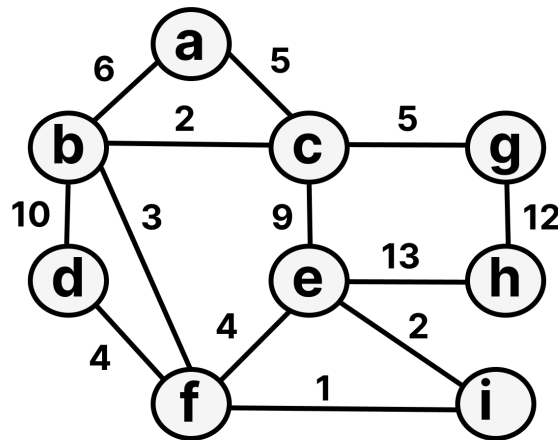
1 Consider the directed graph given below.



- a. **Simulate** a DFS on the graph to find the Discovery and Finish times for each vertex. **3**
CO1 Start with Vertex 'A'.
- b. Now, **simulate** an appropriate algorithm to identify the Strongly Connected **3**
CO1 Components. You can reuse your answer to Question (a) to save time.
- c. **Find** out the minimum number of edges to remove from the graph, so that there exists a **2**
CO1 topological order among the vertices. **Explain** your reasoning.
- d. If we consider each SCC [that you found in Question (b)] as a single vertex, will there **2**
CO1 be any remaining cycles in the graph? **Draw** the graph to justify your answer.

2

A country has several cities that need to be connected by highways. Each edge in the graph represents a potential highway connection between two cities, with weights indicating the **construction cost** of building that highway.



a. The government wants to ensure all cities remain connected while minimizing the total construction cost. Simulate a suitable algorithm to determine which highways should be built. Report the total minimum construction cost. 4

b. Is it possible for a road network (not necessarily the one given above) to have multiple different configurations of highways, each resulting in the same minimum total construction cost? 2

If yes, provide a simple example graph where this occurs. If not, justify your answer with proper reasoning.

c. Now, assume **all highways have been constructed** as shown in the graph. The government has installed toll plazas on each highway. The edge weights now represent the toll fees for using each highway. You're planning a trip from a source city to a destination city and want to minimize your total toll cost. You have a special coupon that allows you to skip paying the toll on **only one** highway (you can use it at most once during your journey). 4

Your friend proposed a modified approach: *First, apply standard Dijkstra's algorithm to find the shortest path from source to destination while keeping track of which edge has the highest toll cost on this path. Then, use the coupon on that edge to skip its toll cost. According to your friend, this will guarantee the minimum total toll cost from source to destination.*

Do you agree that this approach will guarantee the shortest path in all cases, regardless of how the cities are connected, given that all toll fees are positive? If yes, provide proper reasoning. If not, provide a counterexample showing where this approach will fail.

3

You are designing a lossless data compression module for a real-time monitoring system. After analyzing system logs, the following character frequency distribution is obtained:

Character	Frequency
A	28
T	16
L	11
S	9
N	7
K	5
H	3

- a.** Even though all character frequencies are unique, one can obtain different Huffman Trees for above characters. Is this possible? **Explain** whether the bit patterns and the optimality of the code will be affected by this. **2**
- CO1**
- b.** **Simulate** the Huffman Encoding algorithm for the given characters. Construct the huffman tree, generate Huffman code and bits required for each character. (consider left edge as 0, right as 1). **4**
- CO1**
- c.** If you were to send the message “THANKS” with an 8-bit encoding scheme, how many extra bits would have been required compared to Huffman encoding? Ignore the size of the code table. **2**
- CO2**
- d.** “In Huffman Encoding, a higher-frequency symbol always gets a shorter codeword than a lower-frequency symbol.” Is this statement always true? **Explain.** **2**
- CO2**

This is the LCS Table for two strings $X=X_1X_2X_3X_4$ and $Y=Y_1Y_2Y_3Y_4$

$i \backslash j$	-	Y_1	Y_2	Y_3	Y_4
-	0	0	0	0	0
X_1	0	0	0	0	0
X_2	0	0	0	1	1
X_3	0	0	1	1	2
X_4	0	<u>1</u>	1	<u>1</u>	2

- a. What is the direction of cells where values are underlined, i.e which adjacent cells they came from? **CO1** 2
- b. If String $X = \langle \text{Rightmost four digits of Your Student ID} \rangle$, **construct** a correct String Y . **CO1** 2
- c. The formula to find LCS can be implemented both iteratively and recursively. Which one would be more efficient? **Explain** your answer briefly. **CO2** 1